MATHEMATICAL ANALYSIS 1
HOMEWORK 5

(1) For each of the following functions, find all points $x_0 \in \mathbb{R}$ at which f is not defined or not continuous. At each such point compute $\lim_{x\to x_0^-} f(x)$ and $\lim_{x\to x_0^+} f(x)$, and classify the discontinuity (removable, jump, vertical asymptote, or of the second type). Explain your answers.
   (a) $f(x) = \begin{cases} x^2 & \text{for } x \neq 1 \\ 3 & \text{for } x = 1 \end{cases}$
   (b) $f(x) = \operatorname{sign}(x)$
   (c) $f(x) = \frac{x}{x-1}$, $\operatorname{dom}(f) = \mathbb{R} \setminus \{1\}$

(2) Let $f(x) = \lceil x \rceil$ be the ceiling function.
   (a) Prove that f is left-continuous at every $x_0 \in \mathbb{R}$.
   (b) Determine all points at which f is right-continuous, and classify the discontinuities of f.

(3) Prove Proposition 5.5 of the lecture notes:

Proposition. Let $f : \mathbb{R} \to \mathbb{R}$ be defined in a deleted neighborhood of x_0. Then

$$\lim_{x\to x_0} f(x) = L \qquad \Leftrightarrow \qquad \lim_{x\to x_0^+} f(x) = L \quad \text{and} \quad \lim_{x\to x_0^-} f(x) = L$$

where L can be any real number or $\pm\infty$. Moreover, if f is defined at x_0 it is continuous at x_0 if and only if it is both right- and left-continuous at x_0.

(4) Determine the values of $\alpha \in \mathbb{R}$ for which the following functions are continuous on their domains. Explain your answers.
   (a) $f(x) = \begin{cases} \alpha \sin(x + \frac{\pi}{2}) & \text{for } x > 0 \\ 2x^2 + 3 & \text{for } x \leq 0 \end{cases}$
   (b) $f(x) = \begin{cases} 3e^{\alpha x - 1} & \text{for } x \geq 1 \\ x + 2 & \text{for } x < 1 \end{cases}$

(5) Let $f(x) = \sin \frac{1}{x}$, with $\operatorname{dom}(f) = \mathbb{R} \setminus \{0\}$.
   (a) Find all $x \in (0, 1)$ for which $f(x) = 1$, and all $y \in (0, 1)$ for which $f(y) = -1$.
   (b) Using (a) and the definition of the right limit, prove that $\lim_{x\to 0^+} f(x)$ does not exist (it is neither a real number, nor $+\infty$, nor $-\infty$).
   (c) What type of discontinuity does f have at $x_0 = 0$?

(6) Let $x_0 \in \mathbb{R}$ and let f be a monotone increasing function defined on $\mathbb{R} \setminus \{x_0\}$ (it may or may not be defined at x_0). Following the proof of Theorem 5.6, define

$$L_- = \sup_{x<x_0} f(x) \qquad \text{and} \qquad L_+ = \inf_{x>x_0} f(x),$$

which are the left and right limits of f at x_0.
   (a) Explain why L_- and L_+ are real numbers (and not $\pm\infty$).
   (b) Prove that $L_- \leq L_+$. In particular, $L_+ < L_-$ is impossible.
   (c) Prove that if f is also defined at x_0, then $L_- \leq f(x_0) \leq L_+$ (this is Corollary 5.7).

(7) Prove the uniqueness of limits (Theorem 6.1) in the case of a finite point $x_0 \in \mathbb{R}$. There are two cases to prove: *(hint for both cases: by contradiction)*
   (a) If $\lim_{x\to x_0} f(x) = \ell_1$ and $\lim_{x\to x_0} f(x) = \ell_2$ with $\ell_1, \ell_2 \in \mathbb{R}$, then $\ell_1 = \ell_2$.
   (b) It is impossible to have both $\lim_{x\to x_0} f(x) = \ell \in \mathbb{R}$ and $\lim_{x\to x_0} f(x) = +\infty$.

(8) Let $f : \mathbb{R} \to \mathbb{R}$. Prove that if $\lim_{x\to+\infty} f(x) = \ell > 0$ or $\lim_{x\to+\infty} f(x) = +\infty$, then there exists $M > 0$ such that $f(x) > 0$ for all $x > M$ (this is the second item of Theorem 6.2). *Hint: the first item of Theorem 6.2 is proved in the lecture notes.*

(9) Let $f(x) = \frac{x^3-x^2+3}{x^2+3x+2}$.
   (a) Determine $\mathrm{dom}(f)$, and the set on which f is continuous.
   (b) At each real number that does not belong to $\mathrm{dom}(f)$, compute the left and right limits of f. What can you say about the graph of f near these points?

(10) Prove that
$$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}.$$
Hint: multiply both the numerator and the denominator by $1 + \cos x$, and use Example 5.13 of the lecture notes.

(11) In this problem we prove the **Squeeze Theorem** (Theorem 6.9) for a finite point x_0, and three functions f, g, h satisfying $f \le g \le h$ near x_0, with f and h having the limit $\ell \in \mathbb{R}$ as $x \to x_0$ (you are guided in the steps below):
   (a) State the theorem (and state that you shall prove it only in the case $x_0 \in \mathbb{R}$).
   (b) Fix $\varepsilon > 0$.
   (c) With this ε write the definition of what it means that $\lim_{x\to x_0} f(x) = \ell$.
   (d) With the same ε write the definition of what it means that $\lim_{x\to x_0} h(x) = \ell$.
   (e) Using the previous two steps, and the hypothesis $f \le g \le h$ near x_0, find a deleted neighborhood of x_0 (depending on ε) on which $|g(x) - \ell| < \varepsilon$.
   (f) Conclude that, since $\varepsilon > 0$ was arbitrary, the theorem follows by the definition of the limit (applied to g).

(12) Compute the following limits. In each case, explain which results from the lecture notes you use.
   (a) $\lim_{x\to+\infty} \frac{\cos x}{\sqrt{x}}$
   (b) $\lim_{x\to+\infty} \frac{\lfloor x \rfloor}{x}$
   (c) $\lim_{x\to 0} \frac{\sin^2 x}{x}$
   (d) $\lim_{x\to-\infty} x e^{\sin x}$
   (e) $\lim_{x\to+\infty} \left(x^2 + 3\sin x\right)$

Homework 5 Solutions

(1) In each part, at a point x_0 where f coincides, on a whole neighborhood of x_0, with a function that is continuous at x_0, f is also continuous at x_0: continuity at x_0 only involves the values of f near x_0.

(a) If $x_0 \neq 1$, then $f(x) = x^2$ on the neighborhood $I_r(x_0)$ with $r = |x_0 - 1| > 0$, and x^2 is continuous (Proposition 5.4), so f is continuous at x_0. At $x_0 = 1$: on the deleted neighborhood $I_1^*(1)$ we have $f(x) = x^2$, and x^2 is continuous at 1, so

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x) = \lim_{x\to 1} x^2 = 1.$$

Since $f(1) = 3 \neq 1$, f is not continuous at 1, and the discontinuity is **removable** (redefining $f(1) = 1$ makes f continuous).

(b) The sign function is constant on $(-\infty, 0)$ and on $(0, +\infty)$, so it is continuous at every $x_0 \neq 0$ (take $\delta = |x_0|$: for $|x - x_0| < \delta$ we have $f(x) = f(x_0)$). At $x_0 = 0$: for $x \in (0, \delta)$ we have $f(x) = 1$, and for $x \in (-\delta, 0)$ we have $f(x) = -1$, for every $\delta > 0$. Hence

$$\lim_{x\to 0^-} f(x) = -1 \neq 1 = \lim_{x\to 0^+} f(x),$$

so f has a **jump** discontinuity at 0 (and $f(0) = 0$ is neither of the one-sided limits).

(c) f is a rational function, so it is continuous on its domain $\mathbb{R} \setminus \{1\}$ (Corollary 6.7). At $x_0 = 1$ (where f is not defined) the numerator tends to $1 > 0$ and the denominator tends to 0, so we are in the case $\frac{C}{0}$ with $C = 1 > 0$. The denominator $x - 1$ is negative for $x < 1$ and positive for $x > 1$. Hence

$$\lim_{x\to 1^-} \frac{x}{x-1} = -\infty, \qquad \lim_{x\to 1^+} \frac{x}{x-1} = +\infty,$$

and the line $x = 1$ is a **vertical asymptote**. (Directly from the definition: given $A > 0$, let $\delta = \min\{\frac{1}{2}, \frac{1}{2A}\}$. For $1 < x < 1 + \delta$ we have $x > 1$ and $0 < x - 1 < \frac{1}{2A}$, so $\frac{x}{x-1} > \frac{1}{x-1} > 2A > A$. For $1 - \delta < x < 1$ we have $x > \frac{1}{2}$ and $-\frac{1}{2A} < x - 1 < 0$, so, dividing by the negative number $x - 1$, $\frac{x}{x-1} < \frac{1/2}{x-1} < -A$.)

(2) (a) *Proof.* Let $x_0 \in \mathbb{R}$ and let $n = \lceil x_0 \rceil$, so that $n - 1 < x_0 \leq n$ and $f(x_0) = n$. Let $\delta = x_0 - (n-1) > 0$. For every $x \in (x_0 - \delta, x_0) = (n-1, x_0)$ we have $n - 1 < x < x_0 \leq n$, hence $\lceil x \rceil = n$. Therefore, for every $\varepsilon > 0$ and every $x \in (x_0 - \delta, x_0)$,

$$|f(x) - f(x_0)| = |n - n| = 0 < \varepsilon,$$

so $\lim_{x\to x_0^-} f(x) = f(x_0)$, i.e. f is left-continuous at x_0. □

(b) *Non-integer points.* If $x_0 \notin \mathbb{Z}$, let $n = \lceil x_0 \rceil$, so $n - 1 < x_0 < n$. With $\delta = \min\{x_0 - (n-1), n - x_0\} > 0$, every $x \in (x_0 - \delta, x_0 + \delta)$ satisfies $n - 1 < x < n$, so $f(x) = n = f(x_0)$. Hence f is continuous (in particular right-continuous) at x_0.

Integer points. If $x_0 = n \in \mathbb{Z}$, then $f(n) = n$, while for $x \in (n, n+1)$ we have $f(x) = n + 1$. Hence $\lim_{x\to n^+} f(x) = n + 1 \neq n = f(n)$, so f is *not* right-continuous at n. By (a), $\lim_{x\to n^-} f(x) = f(n) = n$.

Therefore f is right-continuous exactly at the points of $\mathbb{R} \setminus \mathbb{Z}$. The discontinuities of f are the integers, and at each integer n the one-sided limits exist and are different (n from the left, $n+1$ from the right): they are **jump** discontinuities.

(3) *Proof.* **Finite L.** ($\Rightarrow$) Suppose $\lim_{x\to x_0} f(x) = L$ and let $\varepsilon > 0$. There is $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - x_0| < \delta$. In particular this holds for all x with $0 < x - x_0 < \delta$ and for all x with $0 < x_0 - x < \delta$. Hence both one-sided limits exist and equal L.

($\Leftarrow$) Suppose both one-sided limits equal L and let $\varepsilon > 0$. There are $\delta_1, \delta_2 > 0$ such that

$$0 < x - x_0 < \delta_1 \;\Rightarrow\; |f(x) - L| < \varepsilon, \qquad 0 < x_0 - x < \delta_2 \;\Rightarrow\; |f(x) - L| < \varepsilon.$$

Let $\delta = \min\{\delta_1, \delta_2\}$. If $0 < |x - x_0| < \delta$, then either $0 < x - x_0 < \delta \leq \delta_1$ or $0 < x_0 - x < \delta \leq \delta_2$; in both cases $|f(x) - L| < \varepsilon$. Hence $\lim_{x\to x_0} f(x) = L$.

$L = +\infty$. The same argument works with the condition $|f(x) - L| < \varepsilon$ replaced by $f(x) > A$ (for a fixed $A > 0$): ($\Rightarrow$) the δ for the two-sided limit works for each side; ($\Leftarrow$) take $\delta = \min\{\delta_1, \delta_2\}$,

where δ_1, δ_2 are given by the right and left limits for this A. The case $L = -\infty$ is identical, with $f(x) < -A$.

Continuity. Suppose f is defined at x_0. Then f is continuous at x_0 if and only if $\lim_{x\to x_0} f(x) = f(x_0)$, which by the first part (with $L = f(x_0)$) holds if and only if $\lim_{x\to x_0^+} f(x) = f(x_0)$ and $\lim_{x\to x_0^-} f(x) = f(x_0)$, i.e. if and only if f is right- and left-continuous at x_0. □

(4) (a) Since $\sin(x + \frac{\pi}{2}) = \sin x \cos\frac{\pi}{2} + \cos x \sin\frac{\pi}{2} = \cos x$, we have $f(x) = \alpha\cos x$ for $x > 0$ and $f(x) = 2x^2 + 3$ for $x \le 0$. Both expressions are continuous on $\mathbb{R}$ (Proposition 5.4 and Corollary 6.6), so f is continuous at every $x_0 \neq 0$ (it coincides with one of them on a neighborhood of x_0). At $x_0 = 0$:

$$f(0) = 3, \qquad \lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} (2x^2 + 3) = 3, \qquad \lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \alpha\cos x = \alpha\cos 0 = \alpha.$$

So f is always left-continuous at 0, and by Proposition 5.5 it is continuous at 0 if and only if $\alpha = 3$. Hence f is continuous on its domain $\mathbb{R}$ if and only if $\alpha = 3$.

(b) For $x \ge 1$ we write $3e^{\alpha x - 1} = \frac{3}{e}(e^\alpha)^x$. If $\alpha \neq 0$ this is a constant times an exponential function with base $e^\alpha > 0$, $e^\alpha \neq 1$; if $\alpha = 0$ it is the constant $\frac{3}{e}$. In both cases $g(x) = 3e^{\alpha x - 1}$ is continuous on $\mathbb{R}$, and so is $x + 2$. Hence f is continuous at every $x_0 \neq 1$. At $x_0 = 1$:

$$f(1) = 3e^{\alpha - 1}, \qquad \lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} g(x) = g(1) = 3e^{\alpha-1}, \qquad \lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} (x + 2) = 3.$$

So f is always right-continuous at 1, and by Proposition 5.5 it is continuous at 1 if and only if $3e^{\alpha-1} = 3$, i.e. $e^{\alpha-1} = 1$, i.e. $\alpha = 1$. Hence f is continuous on $\mathbb{R}$ if and only if $\alpha = 1$.

(5) (a) $f(x) = 1$ if and only if $\frac{1}{x} = \frac{\pi}{2} + 2n\pi$ for some $n \in \mathbb{Z}$. For $x \in (0, 1)$ we need $\frac{1}{x} > 1$, which (since $\frac{\pi}{2} > 1$) holds exactly for $n \in \mathbb{N}$. Hence the solutions in $(0, 1)$ are

$$x_n = \frac{1}{\frac{\pi}{2} + 2n\pi}, \qquad n \in \mathbb{N}.$$

Similarly $f(y) = -1$ if and only if $\frac{1}{y} = \frac{3\pi}{2} + 2n\pi$, and the solutions in $(0, 1)$ are

$$y_n = \frac{1}{\frac{3\pi}{2} + 2n\pi}, \qquad n \in \mathbb{N}.$$

(b) *Proof.* Note that $0 < y_n < x_n < \frac{1}{2n\pi}$ for every $n \ge 1$, so for every $\delta > 0$ there is n (any $n > \frac{1}{2\pi\delta}$) with $x_n, y_n \in (0, \delta)$.
The limit is not a real number. Suppose $\lim_{x\to 0^+} f(x) = \ell \in \mathbb{R}$. Taking $\varepsilon = 1$, there is $\delta > 0$ such that $|f(x) - \ell| < 1$ for all $x \in (0, \delta)$. Choose n with $x_n, y_n \in (0, \delta)$. Then

$$2 = |f(x_n) - f(y_n)| \le |f(x_n) - \ell| + |\ell - f(y_n)| < 1 + 1 = 2,$$

a contradiction.
The limit is not $\pm\infty$. If $\lim_{x\to 0^+} f(x) = +\infty$, then taking $A = 1$ there would be $\delta > 0$ with $f(x) > 1$ for all $x \in (0, \delta)$; but $|f(x)| \le 1$ for all x. Similarly $f(x) < -1$ is impossible, so the limit is not $-\infty$. □

(c) The right limit at 0 does not exist, so 0 is neither a removable discontinuity (the limit does not exist), nor a jump (the one-sided limits do not both exist), nor a vertical asymptote (no one-sided limit is $\pm\infty$). It is a **discontinuity of the second type**. (Since f is odd, the left limit does not exist either.)

(6) (a) The sets $\{f(x) \mid x < x_0\}$ and $\{f(x) \mid x > x_0\}$ are nonempty. Fix any $y > x_0$. For every $x < x_0$ we have $x < y$, so $f(x) \le f(y)$: the first set is bounded from above (by $f(y)$), hence $L_- \in \mathbb{R}$. Similarly, fixing any $x < x_0$, every $y > x_0$ satisfies $f(y) \ge f(x)$, so the second set is bounded from below and $L_+ \in \mathbb{R}$.

(b) *Proof.* By (a), for every $y > x_0$ the number $f(y)$ is an upper bound of $\{f(x) \mid x < x_0\}$. Since L_- is the least upper bound, $L_- \le f(y)$ for every $y > x_0$. Hence L_- is a lower bound of $\{f(y) \mid y > x_0\}$, and since L_+ is the greatest lower bound, $L_- \le L_+$. □

(c) *Proof.* Since f is now defined at x_0, its monotonicity refers to its whole domain, which contains x_0. Hence for every $x < x_0$ we have $f(x) \le f(x_0)$, so $f(x_0)$ is an upper bound of $\{f(x) \mid x <$

$x_0\}$ and therefore $L_- \le f(x_0)$. Similarly, for every $y > x_0$ we have $f(y) \ge f(x_0)$, so $f(x_0)$ is a lower bound of $\{f(y) \mid y > x_0\}$ and $f(x_0) \le L_+$. □

(7) (a) *Proof.* Suppose, by contradiction, that $\ell_1 \neq \ell_2$, and let $\varepsilon = \frac{|\ell_1 - \ell_2|}{2} > 0$. Since $f(x) \to \ell_1$, there is $\delta_1 > 0$ such that $|f(x) - \ell_1| < \varepsilon$ for all $0 < |x - x_0| < \delta_1$; since $f(x) \to \ell_2$, there is $\delta_2 > 0$ such that $|f(x) - \ell_2| < \varepsilon$ for all $0 < |x - x_0| < \delta_2$. Let $\delta = \min\{\delta_1, \delta_2\}$ and take any x with $0 < |x - x_0| < \delta$ (for instance $x = x_0 + \frac{\delta}{2}$). Then

$$|\ell_1 - \ell_2| \le |\ell_1 - f(x)| + |f(x) - \ell_2| < 2\varepsilon = |\ell_1 - \ell_2|,$$

a contradiction. Hence $\ell_1 = \ell_2$. □

Remark. We have tacitly assumed that f is defined on a deleted neighborhood of x_0 (this is part of the definition of $\lim_{x\to x_0} f(x)$), so that, after shrinking δ if necessary, the point $x_0 + \frac{\delta}{2}$ lies in the domain of f.

(b) *Proof.* Suppose, by contradiction, that both hold. Taking $\varepsilon = 1$ in the definition of $\lim_{x\to x_0} f(x) = \ell$, there is $\delta_1 > 0$ such that $f(x) < \ell + 1$ for all $0 < |x - x_0| < \delta_1$. Taking $A = \max\{1, \ell + 1\} > 0$ in the definition of the limit $+\infty$, there is $\delta_2 > 0$ such that $f(x) > A \ge \ell + 1$ for all $0 < |x - x_0| < \delta_2$. For x with $0 < |x - x_0| < \min\{\delta_1, \delta_2\}$ we get $\ell + 1 < f(x) < \ell + 1$, a contradiction. □

(8) *Proof.* *Case* $\lim_{x\to+\infty} f(x) = \ell > 0$. Let $\varepsilon = \frac{\ell}{2} > 0$. By the definition of the limit at $+\infty$ there exists $N_\varepsilon \in \mathbb{R}$ such that $|\ell - f(x)| < \frac{\ell}{2}$ for all $x > N_\varepsilon$, i.e.

$$f(x) \in \left(\ell - \frac{\ell}{2}, \ell + \frac{\ell}{2}\right) = \left(\frac{\ell}{2}, \frac{3\ell}{2}\right) \subset (0, +\infty).$$

Let $M = \max\{N_\varepsilon, 1\} > 0$. Then $f(x) > 0$ for all $x > M$.

Case $\lim_{x\to+\infty} f(x) = +\infty$. Take $A = 1$ in the definition: there exists $N_1 \in \mathbb{R}$ such that $f(x) > 1$ for all $x > N_1$. Let $M = \max\{N_1, 1\} > 0$. Then $f(x) > 1 > 0$ for all $x > M$. □

(9) (a) The denominator factors as $x^2 + 3x + 2 = (x + 1)(x + 2)$, so $\mathrm{dom}(f) = \mathbb{R} \setminus \{-2, -1\}$. Since f is a rational function, it is continuous on its whole domain (Corollary 6.7).

(b) The numerator $p(x) = x^3 - x^2 + 3$ is continuous, with $p(-1) = -1 - 1 + 3 = 1 > 0$ and $p(-2) = -8 - 4 + 3 = -9 < 0$. The denominator $q(x) = (x+1)(x+2)$ tends to 0 at both points, and its sign is

$$q(x) > 0 \text{ for } x < -2, \qquad q(x) < 0 \text{ for } -2 < x < -1, \qquad q(x) > 0 \text{ for } x > -1.$$

So at both points we have an expression of the form $\frac{C}{0}$ with $C \neq 0$, and the sign of the limit is determined by the signs of C and of q near the point:

$$\lim_{x\to-2^-} f(x) = -\infty, \qquad \lim_{x\to-2^+} f(x) = +\infty, \qquad \lim_{x\to-1^-} f(x) = -\infty, \qquad \lim_{x\to-1^+} f(x) = +\infty.$$

(At -2: $C = -9 < 0$, and $q > 0$ to the left, $q < 0$ to the right. At -1: $C = 1 > 0$, and $q < 0$ to the left, $q > 0$ to the right.) Hence the lines $x = -2$ and $x = -1$ are vertical asymptotes of the graph of f; near each of them the graph goes down to $-\infty$ on the left and up to $+\infty$ on the right.

(10) *Proof.* For $0 < |x| < \pi$ we have $\cos x > -1$, so $1 + \cos x > 0$ and we may multiply and divide by it:

$$\frac{1 - \cos x}{x^2} = \frac{(1 - \cos x)(1 + \cos x)}{x^2(1 + \cos x)} = \frac{1 - \cos^2 x}{x^2(1 + \cos x)} = \left(\frac{\sin x}{x}\right)^2 \cdot \frac{1}{1 + \cos x}.$$

By Example 5.13, $\lim_{x\to0} \frac{\sin x}{x} = 1$, so by the Algebra of Limits (Theorem 6.5, product)

$$\lim_{x\to0} \left(\frac{\sin x}{x}\right)^2 = 1.$$

Since cos is continuous, $\lim_{x\to0}(1+\cos x) = 2$, and by the Algebra of Limits (quotient) $\lim_{x\to0} \frac{1}{1+\cos x} = \frac{1}{2}$. Using the Algebra of Limits once more (product),

$$\lim_{x\to0} \frac{1 - \cos x}{x^2} = 1 \cdot \frac{1}{2} = \frac{1}{2}.$$

□

(11) (a) **Theorem (Squeeze Theorem).** Let $f, g, h : \mathbb{R} \to \mathbb{R}$ and let $x_0 \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\}$. If $\lim_{x\to x_0} f(x) = \lim_{x\to x_0} h(x) = \ell \in \mathbb{R}$ and $f \le g \le h$ in a deleted neighborhood of x_0, then $\lim_{x\to x_0} g(x) = \ell$. We prove it only in the case $x_0 \in \mathbb{R}$.

(b) Let $\varepsilon > 0$, and let $\delta_0 > 0$ be such that $f(x) \le g(x) \le h(x)$ for all $x \in I^*_{\delta_0}(x_0)$.

(c) Since $\lim_{x\to x_0} f(x) = \ell$, there exists $\delta_f > 0$ such that for all $0 < |x - x_0| < \delta_f$ we have $|f(x) - \ell| < \varepsilon$, i.e. $\ell - \varepsilon < f(x) < \ell + \varepsilon$.

(d) Since $\lim_{x\to x_0} h(x) = \ell$, there exists $\delta_h > 0$ such that for all $0 < |x - x_0| < \delta_h$ we have $|h(x) - \ell| < \varepsilon$, i.e. $\ell - \varepsilon < h(x) < \ell + \varepsilon$.

(e) Let $\delta = \min\{\delta_0, \delta_f, \delta_h\} > 0$. For every $x \in I^*_\delta(x_0)$, i.e. $0 < |x - x_0| < \delta$, all three conditions hold, so

$$\ell - \varepsilon < f(x) \le g(x) \le h(x) < \ell + \varepsilon,$$

i.e. $|g(x) - \ell| < \varepsilon$.

(f) We have shown that for every $\varepsilon > 0$ there exists $\delta > 0$ such that $|g(x) - \ell| < \varepsilon$ for all $0 < |x - x_0| < \delta$. By the definition of the limit, $\lim_{x\to x_0} g(x) = \ell$.

(12) (a) First, $\lim_{x\to+\infty} \frac{1}{\sqrt{x}} = 0$: given $\varepsilon > 0$, let $N_\varepsilon = \frac{1}{\varepsilon^2}$; for $x > N_\varepsilon$ we have $\sqrt{x} > \frac{1}{\varepsilon}$ (the square root is strictly increasing), so $0 < \frac{1}{\sqrt{x}} < \varepsilon$. Since $|\cos x| \le 1$, the function $\cos x$ is bounded, and by the Squeeze to 0 Theorem (Corollary 6.10),

$$\lim_{x\to+\infty} \frac{\cos x}{\sqrt{x}} = \lim_{x\to+\infty} \cos x \cdot \frac{1}{\sqrt{x}} = 0.$$

(b) For every x we have $x - 1 < \lfloor x \rfloor \le x$, so for $x > 0$,

$$1 - \frac{1}{x} < \frac{\lfloor x \rfloor}{x} \le 1.$$

Since $\lim_{x\to+\infty} \frac{1}{x} = 0$, the Algebra of Limits gives $\lim_{x\to+\infty} \left(1 - \frac{1}{x}\right) = 1$. By the Squeeze Theorem (Theorem 6.9) with $f(x) = 1 - \frac{1}{x}$ and $h(x) = 1$,

$$\lim_{x\to+\infty} \frac{\lfloor x \rfloor}{x} = 1.$$

(c) For $x \neq 0$, $\frac{\sin^2 x}{x} = \sin x \cdot \frac{\sin x}{x}$. Since sin is continuous, $\lim_{x\to 0} \sin x = \sin 0 = 0$, and by Example 5.13 $\lim_{x\to 0} \frac{\sin x}{x} = 1$. By the Algebra of Limits (product),

$$\lim_{x\to 0} \frac{\sin^2 x}{x} = 0 \cdot 1 = 0.$$

(d) Since $\sin x \ge -1$ and the exponential function is increasing, $e^{\sin x} \ge e^{-1}$ for all x. Multiplying by $x < 0$ reverses the inequality: for $x < 0$,

$$xe^{\sin x} \le \frac{x}{e}.$$

Moreover $\lim_{x\to-\infty} \frac{x}{e} = -\infty$ (given $A > 0$, for $x < -eA$ we have $\frac{x}{e} < -A$). By the Squeeze to $\pm\infty$ Theorem (Theorem 6.11, the case $-\infty$),

$$\lim_{x\to-\infty} xe^{\sin x} = -\infty.$$

(e) Since $\sin x \ge -1$, we have $x^2 + 3\sin x \ge x^2 - 3$ for all x, and $\lim_{x\to+\infty}(x^2 - 3) = +\infty$ (given $A > 0$, for $x > \sqrt{A+3}$ we have $x^2 - 3 > A$). By the Squeeze to $\pm\infty$ Theorem (Theorem 6.11),

$$\lim_{x\to+\infty} \left(x^2 + 3\sin x\right) = +\infty.$$